EXPERIMENT :07

Name:Akansha

Create Teacher table with the following fields (Name, DeptNo, Date of joining, DeptName, Location, Salary)

SQL QUERIES:

a) Insert five records

```sql
CREATE DATABASE lab7;
USE lab7;
CREATE TABLE Teacher (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);

INSERT INTO Teacher VALUES
('Amit Sharma', 101, '2020-06-15', 'Mathematics', 'Delhi', 50000),
('Neha Verma', 102, '2019-03-10', 'Commerce', 'Mumbai', 55000),
('Raj Patel', 103, '2021-01-20', 'Mathematics', 'Ahmedabad', 48000),
('Sneha Gupta', 104, '2018-07-25', 'Science', 'Delhi', 60000),
('Vikas Rao', 105, '2022-02-12', 'Commerce', 'Bangalore', 52000);
```

b) **Give Increment of 25% salary for Mathematics Department .**

```sql
START TRANSACTION;

UPDATE Teacher
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics'
LIMIT 100;
```

c) Perform Rollback command

ROLLBACK;

d) GiveIncrementof15%salaryforCommerce Department

e) Perform commit command

```sql
UPDATE Teacher
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce'
LIMIT 100;

COMMIT;
```

EXPERIMENT:08

Create Sales table with the following fields (Sales No, Salesname, Branch, Salesamount, DOB)

a) Insert five records
b) Calculate total salesamount in each branch
c) Calculate average salesamount in each branch
d) Display all the salesmen, DOB who are born in the month of December as day in character format i.e. 21-Dec-09
e) Display the name and DOB of salesman in alphabetical order of the month.

SQL QUERIES AND OUTPUT :

```sql
CREATE DATABASE lab8;
USE lab8;

CREATE TABLE Sales (
    SalesNo INT,
    SalesName VARCHAR(50),
    Branch VARCHAR(50),
    SalesAmount DECIMAL(10,2),
    DOB DATE
);
```

a) Insert five records

```sql
INSERT INTO Sales VALUES
(1, 'Amit', 'Delhi', 5000, '1995-12-21'),
(2, 'Neha', 'Mumbai', 7000, '1998-05-10'),
(3, 'Raj', 'Delhi', 6000, '1992-12-15'),
(4, 'Sneha', 'Chennai', 8000, '1996-07-25'),
(5, 'Vikas', 'Mumbai', 5500, '1994-12-05');
SELECT * FROM Sales;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SalesNo	SalesName	Branch	SalesAmount	DOB
1	Amit	Delhi	5000.00	1995-12-21
2	Neha	Mumbai	7000.00	1998-05-10
3	Raj	Delhi	6000.00	1992-12-15
4	Sneha	Chennai	8000.00	1996-07-25
5	Vikas	Mumbai	5500.00	1994-12-05

Sales 1

b) Calculate total salesamount in each branch

```
SELECT Branch, SUM(SalesAmount) AS Total_Sales
FROM Sales
GROUP BY Branch;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

Branch	Total_Sales
Delhi	11000.00
Mumbai	12500.00
Chennai	8000.00

Result 2

c) Calculate average salesamount in each branch

```
SELECT Branch, AVG(SalesAmount) AS Avg_Sales
FROM Sales
GROUP BY Branch;

SELECT SalesName,
```

Result Grid | Filter Rows: | Export: | Wrap C

Branch	Avg_Sales
Delhi	5500.000000
Mumbai	6250.000000
Chennai	8000.000000

d) Display all the salesmen, DOB who are born in the month of December as day in character format i.e. 21-Dec-09

```
SELECT SalesName,
DATE_FORMAT(DOB, '%d-%b-%y') AS Formatted_DOB
FROM Sales
WHERE MONTH(DOB) = 12;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

SalesName	Formatted_DOB
Amit	21-Dec-95
Raj	15-Dec-92
Vikas	05-Dec-94

Result 4

e) Display the name and DOB of salesman in alphabetical order of the month.

```
SELECT SalesName, DOB
FROM Sales
ORDER BY MONTHNAME(DOB);
```

Result Grid | Filter Rows:

SalesName	DOB
Amit	1995-12-21
Raj	1992-12-15
Vikas	1994-12-05
Sneha	1996-07-25
Neha	1998-05-10

Sales 5

EXPERIMENT:09

Consider the following tables namely “DEPARTMENTS” and “EMPLOYEES” Their schemas are as follows,

Departments (dept no , dept name , dept_location); Employees (emp_id , emp_name , emp_salary,dept_no);

a) Develop a query to grant all privileges of employees table into departments table
b) Develop a query to grant some privileges of employees table into departments table
c) Develop a query to revoke all privileges of employees table from departments table
d) Develop a query to revoke some privileges of employees table from departments table
e) Write a query to implement the save point.

SQL QUERIES AND OUTPUT:

```sql
CREATE DATABASE lab9;
USE lab9;
CREATE TABLE Departments (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);

CREATE TABLE Employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT
);

INSERT INTO Departments VALUES
(101, 'HR', 'Delhi'),
(102, 'Finance', 'Mumbai'),
(103, 'IT', 'Bangalore');

INSERT INTO Employees VALUES
(1, 'Amit', 50000, 101),
(2, 'Neha', 60000, 102),
(3, 'Raj', 55000, 103);
```

a) **Develop a query to grant all privileges of employees table into departments table**

```
CREATE USER user1 IDENTIFIED BY 'password';

GRANT ALL PRIVILEGES
ON Employees
TO user1;
```

b) Develop a query to grant some privileges of employees table into departments table

```
GRANT SELECT, INSERT
ON Employees
TO user1;
```

c) Develop a query to revoke all privileges of employees table from departments table

```
REVOKE ALL PRIVILEGES
ON Employees
FROM user1;
```

d) Develop a query to revoke some privileges of employees table from departments table

```
REVOKE INSERT
ON Employees
FROM user1;
```

e) Write a query to implement the save point.

```
START TRANSACTION;

SAVEPOINT sp1;
SET SQL_SAFE_UPDATES = 0;
UPDATE Employees
SET emp_salary = emp_salary + 5000;
SELECT * FROM Employees;
```

OUTPUT:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dep
1	Amit	50000.00	101
2	Neha	60000.00	102
3	Raj	55000.00	103
NULL	NULL	NULL	NULL

```
ROLLBACK TO sp1;
SELECT * FROM Employees;
COMMIT;
```

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Amit	50000.00	101
2	Neha	60000.00	102
3	Raj	55000.00	103
NULL	NULL	NULL	NULL

EXPERIMENT:10

Using the tables “DEPARTMENTS” and “EMPLOYEES” perform the following queries

a) Display the employee details, departments that the departments are same in both the emp and dept.
b) Display the employee name and Department name by implementing a left outer join.
c) Display the employee name and Department name by implementing a right outer join.
d) Display the details of those who draw the salary greater than the average salary

SQL QUERIES AND OUTPUT:

```sql
CREATE DATABASE lab10;
USE lab10;

CREATE TABLE Departments (
    dept_no INT,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);

CREATE TABLE Employees (
    emp_id INT,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT
);

INSERT INTO Departments VALUES
(101, 'HR', 'Delhi'),
(102, 'Finance', 'Mumbai'),
(103, 'IT', 'Bangalore'),
(104, 'Marketing', 'Chennai');

INSERT INTO Employees VALUES
(1, 'Amit', 50000, 101),
(2, 'Neha', 60000, 102),
(3, 'Raj', 55000, 103),
(4, 'Sneha', 70000, NULL),
(5, 'Vikas', 65000, 105);
```

a) Display the employee details, departments that the departments are same in both the emp and dept.

```
SELECT e.*, d.dept_name, d.dept_location
FROM Employees e
INNER JOIN Departments d
ON e.dept_no = d.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no	dept_name	dept_location
1	Amit	50000.00	101	HR	Delhi
2	Neha	60000.00	102	Finance	Mumbai
3	Raj	55000.00	103	IT	Bangalore

Result 1

b) Display the employee name and Department name by implementing a left outer join.

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_no = d.dept_no;
```

Result Grid | Filter Rows: | Expor

emp_name	dept_name
Amit	HR
Neha	Finance
Raj	IT
Sneha	NULL
Vikas	NULL

Result 2

c) Display the employee name and Department name by implementing a right outer join.

```
SELECT e.emp_name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.dept_no = d.dept_no;
```

Result Grid | Filter Rows: | Expo

emp_name	dept_name
Amit	HR
Neha	Finance
Raj	IT
NULL	Marketing

Result 3

d) Display the details of those who draw the salary greater than the average salary

```
SELECT AVG(emp_salary) FROM Employees;
```

Result Grid | Filter Rows: | Export:

AVG(emp_salary)
60000.000000

```
SELECT emp_id, emp_name, emp_salary
FROM Employees
WHERE emp_salary > (
    SELECT AVG(emp_salary) FROM Employees
);
```

Result Grid | Filter Rows: | Export:

emp_id	emp_name	emp_salary
4	Sneha	70000.00
5	Vikas	65000.00

Employees 5